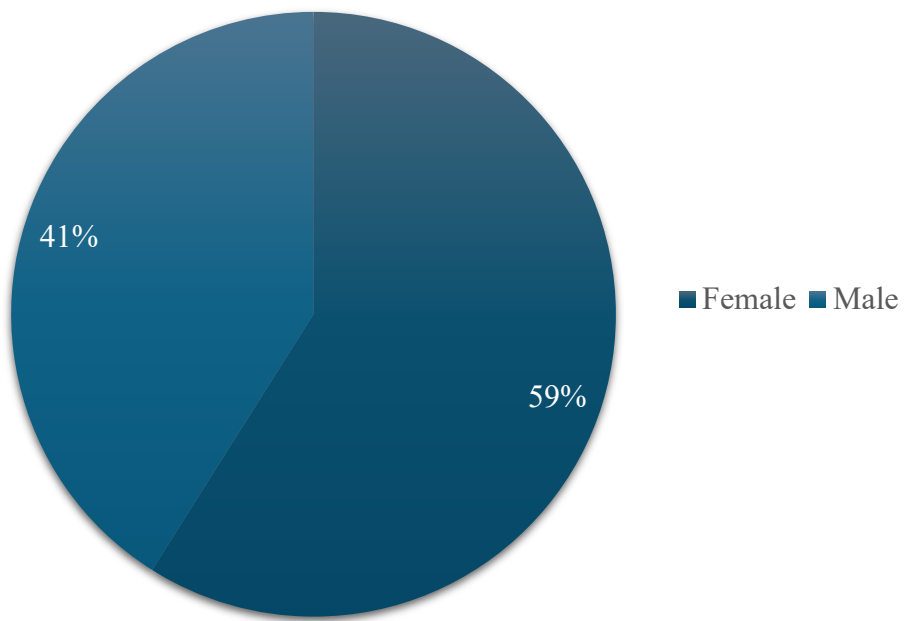


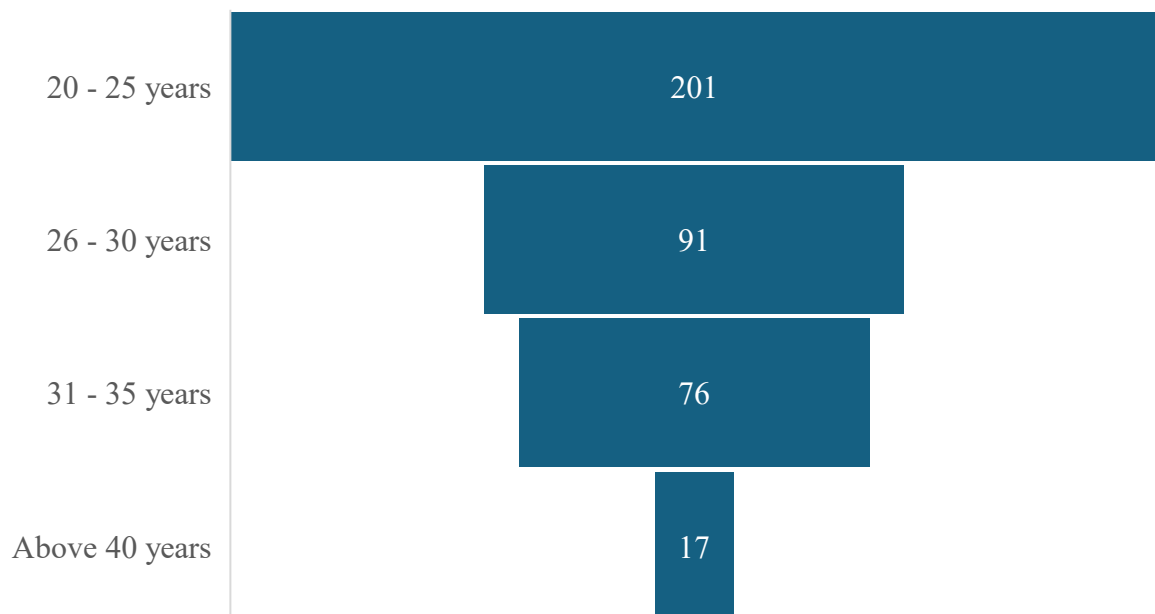
## Demographic Analysis of the Respondents

### Gender



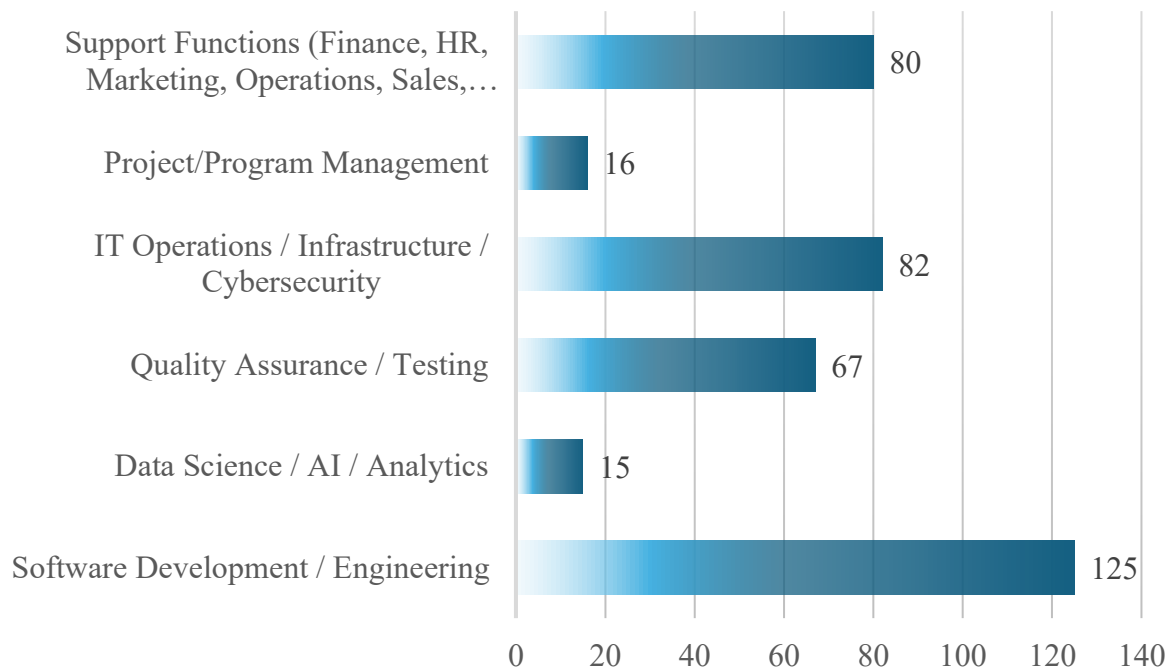
The majority of respondents were females.

### Age



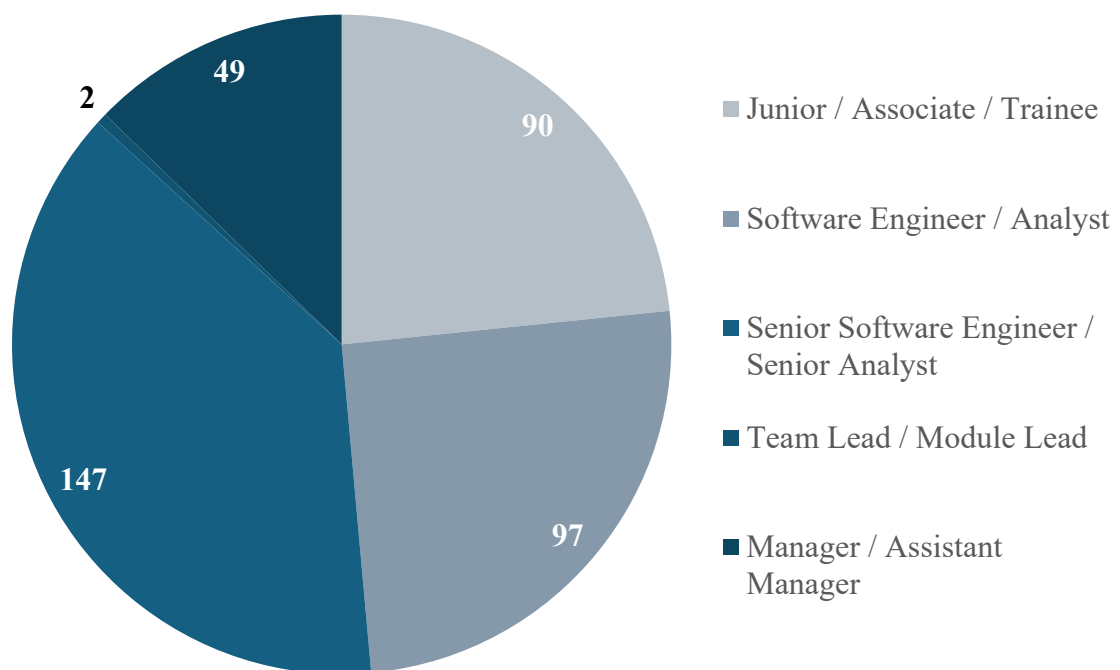
Most respondents fall in the 21–25 years bracket (201), followed by 26–30 years (91) and 31–35 years (76), very few are above 40. The sample is predominantly young professionals.

## Department



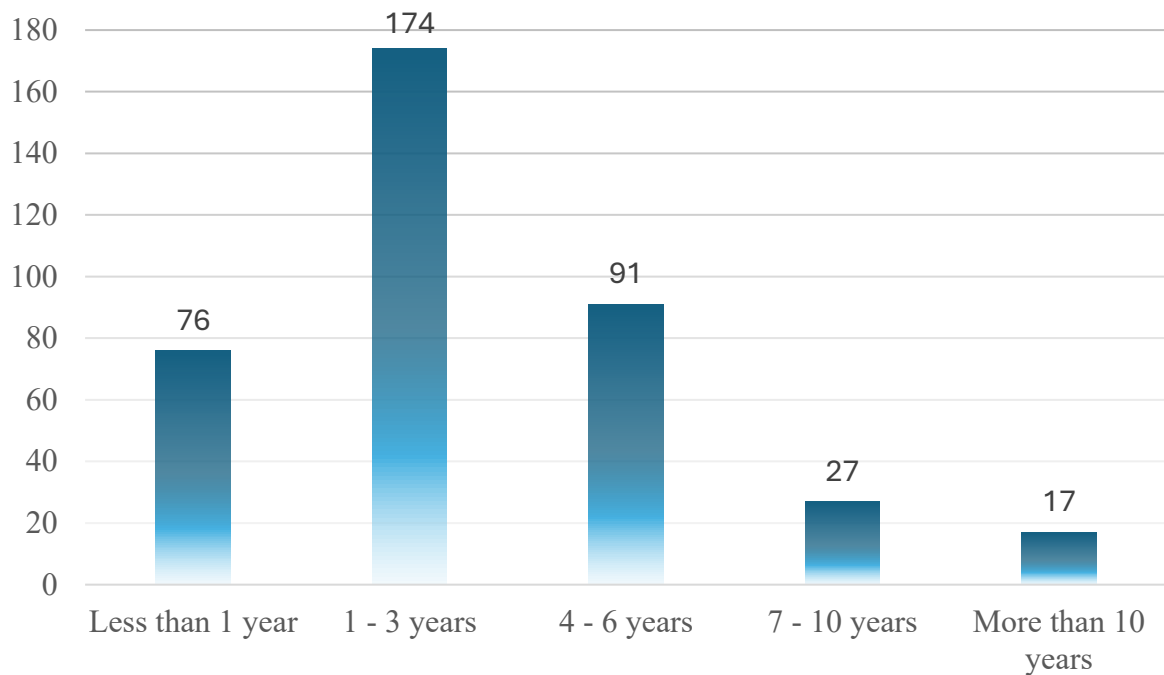
Software Development/Engineering is the largest department, followed closely by Support Functions and IT Operations/Infrastructure/Cybersecurity. Data Science/AI and Project Management have the smallest representation.

## Designation



Senior Software Engineer/Senior Analyst is the most common designation, followed by Software Engineer/Analyst and Junior/Associate/Trainee. Team Leads are the smallest group, and Managers are moderate in number.

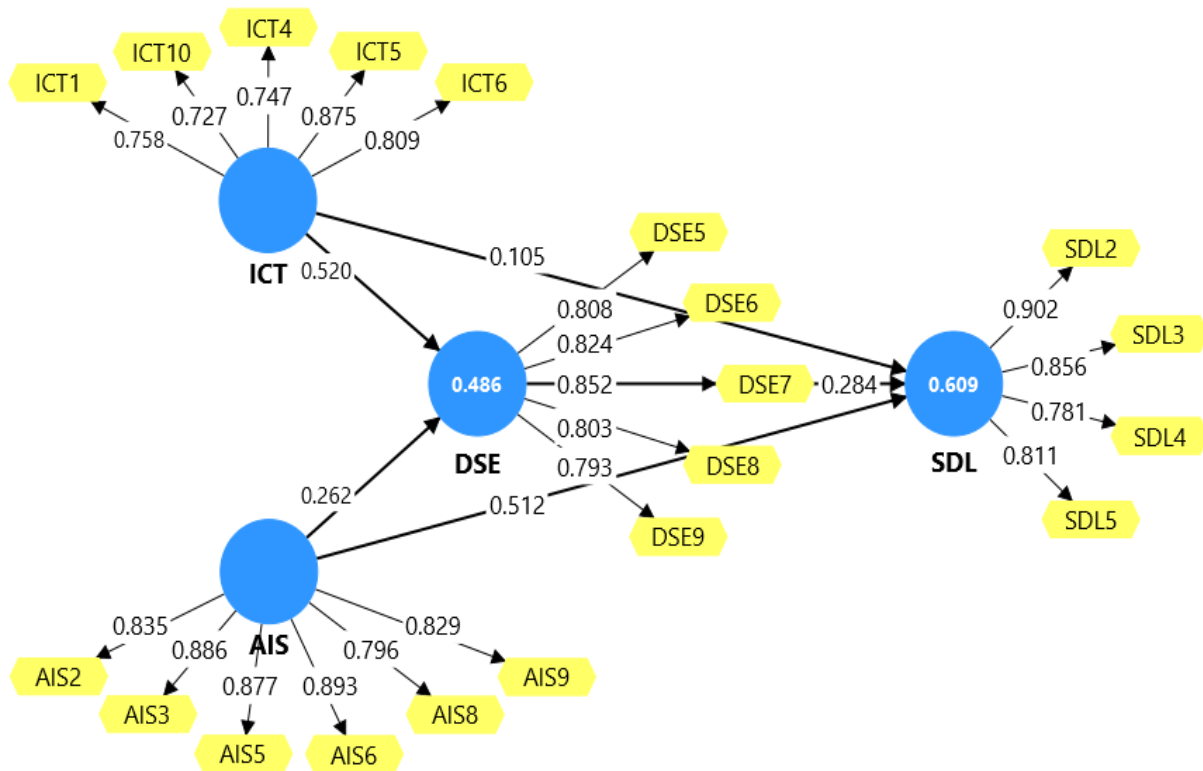
### Years of Corporate Experience



The largest group has 1–3 years of experience, followed by less than 1 year and 4-6 years. This aligns with the young age profile most are early career employees.

## Graphical Output:

### PLS Algorithm



## REFLECTIVE MEASUREMENT MODEL

### Outer Loadings

The reflective measurement model was tested by looking at the outer loadings of all indicators. Outer loadings show how much each question item contributes to the build. During this initial check, some indicators across the constructs recorded values below the minimum acceptable level of 0.70 set by **Hair et al. (2022)**. Items that fall below this level are considered too weak to reliably represent their build and were so removed one at a time. After which all remaining indicators across the five constructs recorded outer loading values above 0.70. As it turns out,

Indicator reliability was checked by squaring each item's outer loading. All items exceeded the minimum level of 0.50 required by **Hair et al. (2022)**, meaning that each item's variance is largely explained by its own build rather than by error.

### Outer Loadings & Reliability and Validity

| Construct                   | Indicator    | Outer loadings | Indicator Reliability | Composite reliability (rho_c) | Average variance extracted (AVE) |
|-----------------------------|--------------|----------------|-----------------------|-------------------------------|----------------------------------|
| AIS<br>(AI Support)         | AIS2 <- AIS  | 0.835          | 0.697                 | 0.941                         | 0.728                            |
|                             | AIS3 <- AIS  | 0.886          | 0.785                 |                               |                                  |
|                             | AIS5 <- AIS  | 0.877          | 0.769                 |                               |                                  |
|                             | AIS6 <- AIS  | 0.893          | 0.797                 |                               |                                  |
|                             | AIS8 <- AIS  | 0.796          | 0.634                 |                               |                                  |
|                             | AIS9 <- AIS  | 0.829          | 0.687                 |                               |                                  |
| DSE (Digital Self-efficacy) | DSE5 <- DSE  | 0.808          | 0.653                 | 0.909                         | 0.666                            |
|                             | DSE6 <- DSE  | 0.824          | 0.679                 |                               |                                  |
|                             | DSE7 <- DSE  | 0.852          | 0.726                 |                               |                                  |
|                             | DSE8 <- DSE  | 0.803          | 0.645                 |                               |                                  |
|                             | DSE9 <- DSE  | 0.793          | 0.629                 |                               |                                  |
| ICT                         | ICT1 <- ICT  | 0.758          | 0.575                 | 0.889                         | 0.616                            |
|                             | ICT10 <- ICT | 0.727          | 0.529                 |                               |                                  |

|                                                              |                       |       |       |       |       |
|--------------------------------------------------------------|-----------------------|-------|-------|-------|-------|
| <b>(Employer<br/>preparedness for<br/>ICT)</b>               | <b>ICT4 &lt;- ICT</b> | 0.747 | 0.558 |       |       |
|                                                              | <b>ICT5 &lt;- ICT</b> | 0.875 | 0.766 |       |       |
|                                                              | <b>ICT6 &lt;- ICT</b> | 0.809 | 0.654 |       |       |
| <b>SDL<br/><br/>(Sustainable<br/>Digital<br/>Leadership)</b> | <b>SDL2 &lt;- SDL</b> | 0.901 | 0.812 | 0.904 | 0.703 |
|                                                              | <b>SDL3 &lt;- SDL</b> | 0.856 | 0.733 |       |       |
|                                                              | <b>SDL4 &lt;- SDL</b> | 0.781 | 0.610 |       |       |
|                                                              | <b>SDL5 &lt;- SDL</b> | 0.811 | 0.658 |       |       |

## Construct Reliability and Validity

Cronbach's alpha and composite reliability values ( $\rho_a$  and  $\rho_c$ ) were checked for all five constructs. Every build exceeded the threshold of 0.70 set by **Henseler et al. (2016)** and the preferred level of 0.80 suggested by **Daskalakis and Mantas (2008)**, confirming strong internal consistency across the board.

Convergent validity was assessed through the Average Variance Extracted (AVE). As it turns out, all five constructs exceeded the minimum AVE of 0.50 required by Ringle et al. (2020) — AIS (0.728), SDL (0.703), DSE (0.666) and ICT (0.616) confirming that each build captures the majority of variance from its own indicators.

## Discriminant Validity

### i. Heterotrait-Monotrait Ratio (HTMT)

The HTMT ratio was used to check whether the constructs are truly distinct from one another. **Hair et al. (2022)** require all values to fall below 0.90. During the initial analysis, some indicators were inflating HTMT values by sharing too much variance across constructs. These items were removed. In the final model, all HTMT values are well below 0.90, showing the clearest separation of any build in the model.

## HTMT

|     | AIS   | DSE   | ICT   | SDL |
|-----|-------|-------|-------|-----|
| AIS |       |       |       |     |
| DSE | 0.596 |       |       |     |
| ICT | 0.583 | 0.725 |       |     |
| SDL | 0.787 | 0.731 | 0.643 |     |

### ii. Fornell-Larcker Criterion

The Fornell-Larcker Criterion requires the square root of each construct's AVE shown along the diagonal to exceed all off-diagonal inter-construct correlations. As it turns out, before the cross-loading misaligned items were removed, some build pairs did not meet this requirement cleanly. After removal, the diagonal values, AIS (**0.853**), DSE (**0.816**), ICT (**0.785**), and SDL (**0.839**) clearly exceeded all off-diagonal values, confirming discriminant validity at the build level.

#### Fornell-Larcker Criterion

|     | AIS   | DSE   | ICT   | SDL   |
|-----|-------|-------|-------|-------|
| AIS | 0.853 |       |       |       |
| DSE | 0.542 | 0.816 |       |       |
| ICT | 0.538 | 0.662 | 0.785 |       |
| SDL | 0.723 | 0.631 | 0.568 | 0.839 |

### iii. Cross Loadings

The cross-loadings matrix was examined to confirm that each item loads most strongly on the build it is meant to measure, while showing clearly lower values on all other constructs. During this stage, an indicator was found to be misaligned, which was removed to protect the clarity and integrity of the measurement structure. The removal of these items also had a direct corrective effect on the Fornell-Larcker and HTMT results, as discussed above, since the inflated cross-construct variance was the root cause of the earlier violations in those two tests.

### Cross Loadings

|              | <b>AIS</b> | <b>DSE</b> | <b>ICT</b> | <b>SDL</b> |
|--------------|------------|------------|------------|------------|
| <b>AIS2</b>  | 0.835      | 0.323      | 0.377      | 0.639      |
| <b>AIS3</b>  | 0.886      | 0.558      | 0.464      | 0.708      |
| <b>AIS5</b>  | 0.877      | 0.503      | 0.499      | 0.620      |
| <b>AIS6</b>  | 0.893      | 0.528      | 0.514      | 0.636      |
| <b>AIS8</b>  | 0.796      | 0.429      | 0.410      | 0.531      |
| <b>AIS9</b>  | 0.829      | 0.397      | 0.482      | 0.541      |
| <b>DSE5</b>  | 0.599      | 0.808      | 0.394      | 0.521      |
| <b>DSE6</b>  | 0.343      | 0.824      | 0.639      | 0.635      |
| <b>DSE7</b>  | 0.498      | 0.852      | 0.682      | 0.477      |
| <b>DSE8</b>  | 0.383      | 0.803      | 0.449      | 0.483      |
| <b>DSE9</b>  | 0.397      | 0.793      | 0.490      | 0.440      |
| <b>ICT1</b>  | 0.299      | 0.478      | 0.758      | 0.381      |
| <b>ICT10</b> | 0.353      | 0.408      | 0.727      | 0.368      |
| <b>ICT4</b>  | 0.375      | 0.418      | 0.747      | 0.356      |
| <b>ICT5</b>  | 0.604      | 0.736      | 0.875      | 0.604      |
| <b>ICT6</b>  | 0.395      | 0.451      | 0.809      | 0.446      |
| <b>SDL2</b>  | 0.731      | 0.472      | 0.585      | 0.901      |
| <b>SDL3</b>  | 0.470      | 0.521      | 0.478      | 0.856      |
| <b>SDL4</b>  | 0.432      | 0.681      | 0.475      | 0.781      |
| <b>SDL5</b>  | 0.734      | 0.470      | 0.366      | 0.811      |

In the final cross-loadings matrix, all retained items show strong primary loadings on their own constructs and noticeably weaker correlations with the remaining ones. This pattern confirms both convergent and discriminant validity at the item level.

### R Square (R<sup>2</sup>)

|            | <b>R-square</b> |
|------------|-----------------|
| <b>SDL</b> | 0.609           |



SDL has an R-square of 0.609, meaning 60.9% moderate to substantial relationship. Simply put, the value reflects a meaningful level of explanatory power for behavioural research (**Hair et al., 2022**).

## **STRUCTURAL MODEL**

### **Collinearity Statistics (VIF)**

The structural model was assessed by first checking for multicollinearity using the Variance Inflation Factor (VIF) at both the indicator level (outer model) and the build level (inner model).

#### **i. Outer Model:**

Some indicators showed elevated VIF values during the outer model check, suggesting they were duplicating what other items in the same build were already measuring. These were removed following **Hair et al. (2022)**. In the final model, all VIF values are within acceptable limits, confirming that multicollinearity is not a concern.

#### **VIF Outer Model**

|              | <b>VIF</b> |
|--------------|------------|
| <b>AIS2</b>  | 3.243      |
| <b>AIS3</b>  | 3.726      |
| <b>AIS5</b>  | 3.359      |
| <b>AIS6</b>  | 3.503      |
| <b>AIS8</b>  | 2.146      |
| <b>AIS9</b>  | 3.362      |
| <b>DSE5</b>  | 2.074      |
| <b>DSE6</b>  | 2.002      |
| <b>DSE7</b>  | 2.239      |
| <b>DSE8</b>  | 2.011      |
| <b>DSE9</b>  | 1.930      |
| <b>ICT1</b>  | 1.663      |
| <b>ICT10</b> | 1.603      |
| <b>ICT4</b>  | 1.690      |

|             |       |
|-------------|-------|
| <b>ICT5</b> | 2.128 |
| <b>ICT6</b> | 1.958 |
| <b>SDL2</b> | 3.381 |
| <b>SDL3</b> | 3.234 |
| <b>SDL4</b> | 2.159 |
| <b>SDL5</b> | 2.153 |

## ii. Inner Model:

The inner model VIF values check for collinearity among the predictor constructs themselves in relation to the endogenous variables

### VIF Outer Model

|                      | <b>VIF</b> |
|----------------------|------------|
| <b>AIS -&gt; DSE</b> | 1.418      |
| <b>AIS -&gt; SDL</b> | 1.549      |
| <b>DSE -&gt; SDL</b> | 1.950      |
| <b>ICT -&gt; DSE</b> | 1.409      |
| <b>ICT -&gt; SDL</b> | 1.935      |

All values in the final model are comfortably below the 5.0 threshold. These inner model results confirm that no problematic collinearity exists among the predictor constructs, and that the structural path estimates produced by SmartPLS-SEM are stable and trustworthy.

## Path Coefficients

| <b>Path</b>          | <b>Path Coefficients</b> | <b>Standard Deviation</b> | <b>T statistics</b> | <b>P values</b> | <b>Bias</b>         |                      | <b>Result</b> |
|----------------------|--------------------------|---------------------------|---------------------|-----------------|---------------------|----------------------|---------------|
|                      |                          |                           |                     |                 | <b>(Lower) 2.5%</b> | <b>(Upper) 97.5%</b> |               |
| <b>ICT -&gt; DSE</b> | 0.519                    | 0.041                     | 12.690              | 0.000           | 0.435               | 0.594                | Supported     |
| <b>ICT -&gt; SDL</b> | 0.105                    | 0.048                     | 2.192               | 0.028           | 0.011               | 0.198                | Supported     |

|                                        |       |       |        |       |       |       |           |
|----------------------------------------|-------|-------|--------|-------|-------|-------|-----------|
| <b>AIS -&gt;<br/>DSE</b>               | 0.259 | 0.049 | 5.341  | 0.000 | 0.162 | 0.353 | Supported |
| <b>AIS -&gt;<br/>SDL</b>               | 0.512 | 0.038 | 13.568 | 0.000 | 0.442 | 0.590 | Supported |
| <b>DSE -&gt;<br/>SDL</b>               | 0.285 | 0.045 | 6.363  | 0.000 | 0.193 | 0.370 | Supported |
| <b>ICT -&gt;<br/>DSE -&gt;<br/>SDL</b> | 0.148 | 0.028 | 5.333  | 0.000 | 0.096 | 0.205 | Supported |
| <b>AIS -&gt;<br/>DSE -&gt;<br/>SDL</b> | 0.074 | 0.016 | 4.486  | 0.000 | 0.045 | 0.111 | Supported |

### **H1: ICT → SDL: Supported**

Employer Preparedness for ICT has a small but real positive effect on Sustainable Digital Leadership. Organisations that are better prepared with ICT do contribute to stronger digital leadership, though this effect works more as a supporting condition than a primary driver.

### **H2: ICT → DSE: Supported**

Employer Preparedness for ICT is the strongest predictor of Digital Self-Efficacy. When organisations invest in ICT infrastructure and support, employees feel notably more confident in their digital abilities.

### **H2: AIS → SDL: Supported**

AI Support is the strongest direct driver of Sustainable Digital Leadership in the entire model. When employees work meaningfully with AI tools, it has a powerful and clear positive effect on how they lead in a sustainable digital way.

### **H4: AIS → DSE: Supported**

AI Support positively and significantly builds Digital Self-Efficacy. Regular and supported engagement with AI tools helps employees feel more capable and assured in handling digital work.

**H5: DSE → SDL: Supported**

Digital Self-Efficacy positively influences Sustainable Digital Leadership. Employees who feel more confident in their digital capabilities are more likely to think and act as sustainable digital leaders.

**H6: ICT → DSE → SDL: Supported**

Digital Self-Efficacy mediates the relationship between Employer Preparedness for ICT and Sustainable Digital Leadership. ICT preparedness first builds employee digital confidence, and that confidence then drives sustainable digital leadership. The mediation path is significant.

**H7: AIS → DSE → SDL: Supported**

Digital Self-Efficacy mediates the relationship between AI Support and Sustainable Digital Leadership. Part of the reason AI Support improves leadership sustainability is because it first strengthens how confident employees feel about their digital work. This indirect path is statistically significant.